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# Tensor Decomposer

**Presented By**

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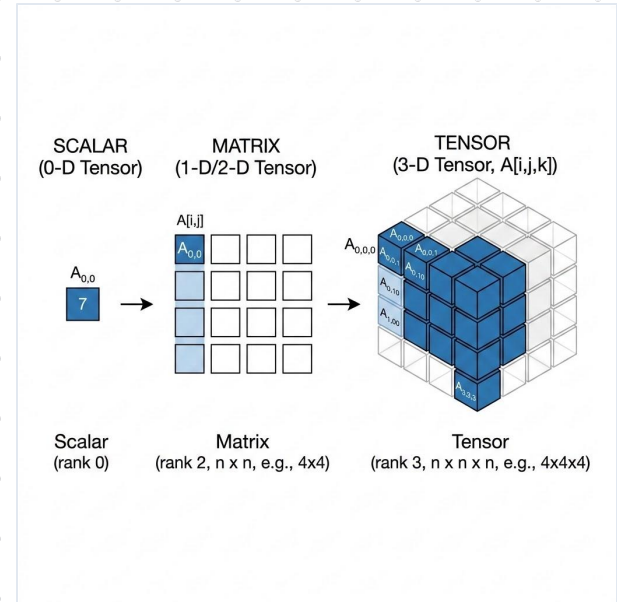
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# Tensors: Multidimensional Data Arrays

- **Mathematical Framework:** Generalization of scalars (0D), vectors (1D), and matrices (2D).
- **N-Dimensional Arrays:** Represent complex, multi-way relationships across multiple axes.
- **Common Applications:** RGB images, Multi feature data representation, and Machine Learning.



# Tensor Decomposition

## 1. Tensor Decomposition

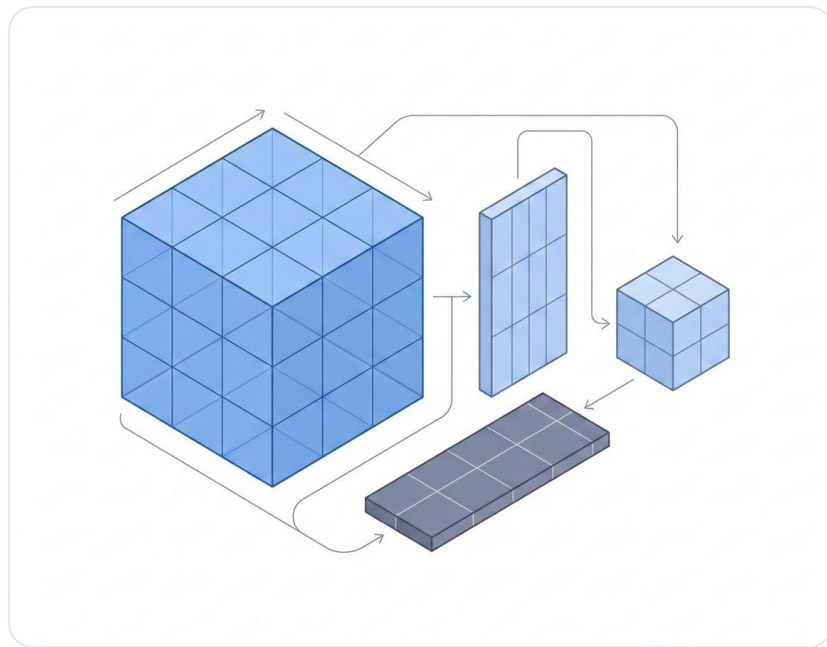
Decomposes complex multi-way tensors into smaller, manageable tensors and factor matrices.

## 2. Feature Extraction

Extracts latent features, reduces dimensionality, and reveals core underlying patterns.

## 3. Practical Applications

Widely applied in data mining, signal noise reduction, image analysis, and compression.



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# Present Challenges of Decomposition

- High computation costs
- No web app available
- Tools are scattered across different places
- Lacks modern optimization features
- Hard to visualize

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# Proposed Solution

## Tensor Decomposer

- Web app to run tensor operations
- Interactive visual tool to see results
- Performance analysis

## tensordecomp

- Python library for offline work
- Supports multiple solution methods
- Includes modern, new techniques

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# Functional Requirements

## 01. Tensor Input

Web interface for tensor input manually or via file

## 02. Decomposition

Multi-method tensor decomposition algorithms.

## 03. Benchmarking

Performance and accuracy benchmarking across methods.

## 04. Error Analysis

Tensor reconstruction and detailed analytical error measurement.

## 05. Visualization

Interactive output presentation and graphical rendering.

## 06. Method Comparison

Comparative analysis evaluate results of various techniques.

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# Non-Functional Requirements

## Puzzle Tensor Optimization

- Specialized optimization routines tailored for complex tensor algorithms.
- Ensures efficiency and reduced computational overhead during execution.

## Offline Python Library

- Standalone tensordecomp package for local environment workflows.
- Fully functional offline operation without external network dependencies.

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# Tools & Technologies



**Python**



**Vercel**



**GitHub**

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# Tensor Decomposer: Web App

Live URL: <https://td4.vercel.app/>

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# tensordecomp: Python Library

**Library:** <https://pypi.org/project/tensordecomp/>

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# Challenges

- **Implementing different algorithms**
- **Visualizing separate outputs**
- **Building and publishing library**

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# Future Scope

- **3D Visualization**
- **Implementing New Methods**
- **Web App Storage**

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# References

- **Kolda, T. G., & Bader, B. W. (2009).** Tensor decompositions and applications. SIAM Review, 51(3), 455–500.
- **Oseledets, I. V. (2011).** Tensor-train decomposition. SIAM Journal on Scientific Computing, 33(5), 2295–2317.
- **Park, Y.-C., Kim, K., & Kang, U. (2025).** PuzzleTensor: A method-agnostic data transformation for compact tensor factorization.



**Thank You**